ZIAD ALI FATHY MOHAMED

AI and Machine Learning Engineer Intern | Python | SQL

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PROFESSIONAL SUMMARY

Results-driven Computer and Information Sciences student specializing in Artificial Intelligence and Machine Learning, with proven technical competencies in predictive modeling, data pipeline automation, and distributed Big Data infrastructure. Completed 300+ hours of advanced training across Creativa, Huawei ICT Academy, and NTI. Proficient in Python, SQL, and Scikit-learn, with hands-on practice building end-to-end Machine Learning workflows and Generative AI applications. Seeking an AI / Machine Learning Internship to deliver scalable, .high-performance data solutions

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL
- **Data Science & Analytics:** Pandas, NumPy, Matplotlib, Exploratory Data Analysis (EDA), Data Cleaning, Feature Engineering
- **Machine Learning & AI:** Scikit-learn, Supervised Learning, Unsupervised Learning, Hyperparameter Tuning, Model Evaluation, Generative AI
- **Databases & Data Stores:** PostgreSQL, MySQL, SQL Server, Git/GitHub
- **Big Data & Distributed Systems:** Apache Hadoop, Apache Spark, Hive, HBase, Flink

WORK EXPERIENCE

Generative AI Engineer Trainee | Echo Inc.

October 2026 – Present

- Engineer 5+ production-grade Generative AI application components and automated prompt pipelines across 4 sprint milestones.
- Author structured technical documentation and automated validation benchmarks for large language model integration workflows.
- Benchmark model inference latency across 3 LLM architectures, reducing token consumption costs by 18%.

Machine Learning Trainee | Creativa

August 2026 – October 2026

- Architected end-to-end Machine Learning pipelines processing 10,000+ data samples with comprehensive cleaning and feature selection.
- Trained and optimized classification and regression models using Scikit-learn, boosting baseline accuracy by 12% to an 88% precision score.
- Evaluated 4 distinct supervised learning algorithms using ROC-AUC and F1-score metrics to determine optimal deployment candidates.

Data Analysis Trainee | Voltex

September 2026 – October 2026

- Queried and aggregated relational datasets exceeding 50,000 records using advanced SQL, delivering 10+ executive KPI metrics.
- Formulated exploratory data analysis (EDA) workflows in Python, accelerating recurring preprocessing cycles by 20% via automation scripts.
- Constructed 8 automated visual reports using Matplotlib and Pandas to monitor data distributions and identify anomaly patterns.

Big Data Associate Trainee | Huawei ICT Academy / NTI

July 2026 – September 2026

- Completed 100+ hours of rigorous technical training on distributed computing architectures, cluster scalability, and enterprise storage.
- Deployed distributed queries across multi-node clusters using Spark and Hive, decreasing batch processing execution latency by 15%.
- Executed MapReduce and HiveQL batch jobs on 500MB+ datasets, verifying 100% data partition integrity across HDFS storage.

KEY PROJECTS

Healthcare & Clinical Data Analysis Pipeline | Python, Pandas, Scikit-learn, SQL

- Automated end-to-end data cleaning, imputation, and normalization protocols across 15,000+ clinical records and 20+ medical variables.
- Developed predictive risk-classification models achieving 89% accuracy validated through rigorous 5-fold cross-validation.
- Programmed automated SQL ingestion pipelines, reducing manual reporting overhead and report preparation turnaround time by 30%.

Enterprise Database & CRUD Analytics Management System | Python, SQL, PostgreSQL/MySQL

- Designed a full-stack data management application handling 20+ relational entities with full ACID transaction compliance.

- Optimized query throughput and execution speed by 25% utilizing strategic B-tree indexing, parameterized queries, and defensive schema rules.

EDUCATION

B.Sc. in Computer and Information Sciences

Benha National University | Expected Graduation: June 2028

Track: Artificial Intelligence & Computer Science

CERTIFICATIONS

- **HCIA-AI V4.0** – Huawei ICT Academy
- **HCIA-Big Data V3.5** – Huawei ICT Academy
- **Huawei Big Data Associate** – NTI / Egyptian Talent Academy
- **AI Fundamentals and Data Analysis** – Skills Area / Partner Programs

LANGUAGES

- **Arabic:** Native
- **English:** Professional Working Proficiency